

Harishlal M E

3rd Year Undergraduate
Computer Science and Engineering

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EDUCATION

SRM Institute of Science and Technology, Chennai (2024 – 2028)

- Bachelor of Technology, Computer Science and Engineering
- Major Degree — Current Cumulative GPA: 9.50/10
- Minor Degree in Artificial Intelligence & Machine Learning — Current Cumulative GPA: 9.50/10 (3-semester specialisation programme, concluding 2026)

EXPERIENCE

SRM Institute of Science and Technology, Chennai (Nov 2025 – Jan 2026)
Undergraduate Research Assistant (UROP) | Advisor: Dr. G. Balamurugan

- Researching safety-critical NLP systems for content moderation, designing recall-focused evaluation pipelines that penalise false negatives over headline accuracy.
- Fine-tuned and calibrated transformer classifiers and authored the resulting manuscript, currently under review at *Computers in Human Behavior* (Elsevier).

Innovexis (backed by IntelliHire Staffing Solution Pvt. Ltd.) (Jan – Apr 2026)
AI/ML Intern

- Delivered three industry machine learning engagements end to end, covering data cleaning, exploratory analysis, feature engineering, hyperparameter tuning and model evaluation in Python.
- Scope: Tesla Stock Price Prediction (SimpleRNN & LSTM) · Real Estate Investment Advisor (Streamlit, MLflow) · PhonePe Transaction Insights (SQL analytics and visualisation).

Google for Developers & AICTE (Jan – Mar 2025)
AI-ML Virtual Internship

- Completed a 10-week intensive programme in ML model development, evaluation and deployment best practices.

PROJECTS

APG-ASR: Gated CNN-Transformer Detector with Agentic Inference (Source Code) 2026 – Present

- Engineered a hybrid CNN-Transformer object detector with gated feature fusion capping transformer influence at ~3.5% of feature magnitude, developed as the minor project for the C-DAC (MeitY) AI & ML certification programme.
- Eliminated gradient collapse in the naive hybrid, restoring stable training from 0.31 → 0.33 mAP@50 on the full COCO 2017 validation set (5,000 images).
- Architected an agentic propose-filter-recheck inference layer that lifts precision by 33% (0.45 → 0.60) while reprocessing only ~5% of spatial regions, holding latency under 100 ms per image.
- Languages and frameworks used: Python, PyTorch, YOLOv8, Transformers.

The Last CEO: AI-Powered Business Strategy Simulator (Source Code) · **Best Project, SRM Insider** June 2026

- Led backend and ML architecture for a quarterly business-strategy simulation in which every player decision is scored by a live XGBoost prediction engine, guiding the player through AI transformation to 2035.
- Layered SHAP-based explainability and an LLM strategic advisor (Groq Llama 3.1) over the prediction engine to justify each recommendation.
- Deployed the FastAPI backend to Render with a CORS-enabled, zero-configuration setup; frontend built in React and TypeScript over SQLAlchemy models.

SpeakSense AI: Real-Time AI Language Coaching Platform (Source Code) · **Hackathon Winner** March 2026

- Conceived, built and shipped a working conversational coaching product in under 24 hours, winning first place at the SRMIST Symposium on Responsible AI hackathon.
- Automated proficiency-level detection and LLM-driven scenario coaching across four modes — job interview, negotiation, public speaking and casual chat.
- Enforced JSON-structured prompting to deliver "Good / Better / Best" rewrites, targeted phonetic pronunciation tips and per-session grammar, fluency and pronunciation scoring.
- Languages and frameworks used: React (Vite), FastAPI, SQLite, Groq API.

Cyberbullying Detection System with BERT (Source Code) December 2025

- Fine-tuned BERT to 94.5% recall and 94.19% F1-score, optimising explicitly against false negatives for a safety-critical moderation setting.
- Calibrated decision thresholds and ran bias analysis, then deployed the model as a Streamlit web application; the work was extended into a journal manuscript.
- Languages and frameworks used: Python, PyTorch, Hugging Face Transformers, Streamlit.

CampusSafe: Campus Lost & Found Management System (Source Code) February 2026

- Built a full-stack platform on a normalised 11-table MySQL schema with a JWT-secured REST API and role-based access control for students and administrators.
- Engineered a Jaccard-similarity matching engine (40–95% confidence scoring) with admin claim workflows and auto-generated institutional PDF reports.
- Languages and frameworks used: React 18, Node.js, Express.js, MySQL 8.0, JWT, PDFKit, Tailwind CSS.

PUBLICATIONS

Recall-Focused Cyberbullying Detection using Transformer Models (2026) Under Review
Computers in Human Behavior (Elsevier)

Authored a manuscript on recall-oriented evaluation of transformer classifiers for content moderation, arguing for false-negative suppression as the primary objective in safety-critical detection. Conducted under the UROP programme at SRM IST.

TECHNICAL SKILLS

Programming Languages

- Python, Java and C.

Technologies

- Machine Learning algorithms with **scikit-learn** and **XGBoost**; feature engineering, hyperparameter tuning and model evaluation.
- Deep Learning, NLP and Computer Vision with **PyTorch**, **TensorFlow**, **Hugging Face Transformers**, **BERT**, **YOLOv8** and **SHAP**.
- Backend and Web: **FastAPI**, **React**, **Node.js**, **Express.js**, **REST APIs**, **Tailwind CSS**, **MySQL** and **SQLite**.
- Tools: **Git**, **Docker**, **Streamlit**, **MLflow** and **Render**.

CERTIFICATIONS

1. **Machine Learning Specialization** | **DeepLearning.AI & Stanford Online** | Coursera (Dec 2025)
2. **Building RAG Agents with LLMs** | **NVIDIA**, Certificate of Competency (Mar 2026)
3. **Oracle Certified Foundations Associate — Agentic AI** | **Oracle University** (Aug 2026)
4. **Certification Course in Artificial Intelligence & Machine Learning** | 400-hour project-based programme, **C-DAC MEPZ**, **MeitY**, **Government of India**
5. **Fundamentals of Object Oriented Programming (Elite, Top 1%)** | **NPTEL**, **IIT Roorkee** (Apr 2025)
6. **AI-ML Virtual Internship** | **Google for Developers & AICTE** (Mar 2025)

COURSEWORK

Major Degree — Computer Science and Engineering

Data Structures and Algorithms · Design and Analysis of Algorithms · Advanced Programming Practices · Object Oriented Design and Programming · Operating Systems · Computer Organization and Architecture · Probability and Queuing Theory · Advanced Calculus and Complex Analysis · Transforms and Boundary Value Problems

Minor Degree — Artificial Intelligence & Machine Learning

Data Science · Machine Learning · Deep Learning · Prompt Engineering · Explainable AI (XAI) · Image Processing

ACHIEVEMENTS

1st Place | Hackathon, Symposium on Responsible AI | SRMIST (Mar 2026)

- Won first place among competing teams at the hackathon conducted by the Department of Computing Technologies in association with the SRM ACM Student Chapter, for SpeakSense AI built inside the 24-hour window.

Best Project & Rank 3 | Overall Individual Rankings, SRM Insider (Jun 2026)

- "Code & Capital · The Last CEO" selected as the club's best project; placed in the top 3 of 30+ members in the overall individual rankings.

Elite + Topper (Top 1%) | NPTEL, IIT Roorkee (Apr 2025)

- Scored 94% in Fundamentals of Object Oriented Programming among 5,246 certified candidates nationally.

1st Prize | "Reuse and Remodel Product", SRMIST (Nov 2024)

- Won first prize in the event organised by the Department of Mechanical Engineering; active participant across multiple inter- and intra-college hackathons and ideathons.

POSITIONS OF RESPONSIBILITY

1. AI/ML Technical Member | SRM Insider (2025 – Present)

- Delivered 3+ AI/ML projects across more than a year with the club's technical wing, contributing to build sprints and community technical events.
- Ranked 3rd overall among 30+ members and awarded Best Project for The Last CEO.

2. Community Outreach Volunteer | Community Connect, SRM IST (Jun 2026)

- Delivered 21 hours of community service at a higher secondary school in Madurai, conducting AI and technology awareness sessions for Classes X-XII on the responsible use of AI for learning and career growth.

LANGUAGES & INTERESTS

Languages: English, Tamil and Sourashtra — fluent | Hindi — conversational | French — basic (subject topper at higher secondary level)

Interests: Applied and responsible AI research · Open-source development · Rapid prototyping in hackathon settings · Creative writing — author of the short story "The Unwelcomed Ride"